

HAT-P-18b

R-2286

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_140220 · created 2026-08-17T00:32:11+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456708.93028 +/- 0.00089 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.183 +/- 0.052
Transit depth (Rp/Rs)^2	3.35 +/- 1.91 [%]
Orbital Inclination (inc)	87.28 +/- 1.87
Airmass coefficient 1 (a1)	0.54 +/- 0.35
Airmass coefficient 2 (a2)	0.14 +/- 0.11
Scatter in the residuals of the lightcurve fit is	67.78 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.75
Transit Duration (day)	0.079 +/- 0.045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.183 ± 0.052 vs 0.1356 (deviation 0.87σ)
Duration fit vs published	0.079 d vs 0.1129 d (deviation 0.72σ)
Mid-time O–C	1.0 min
Depth SNR	3.4
Residual scatter	67.78 %

Light curve

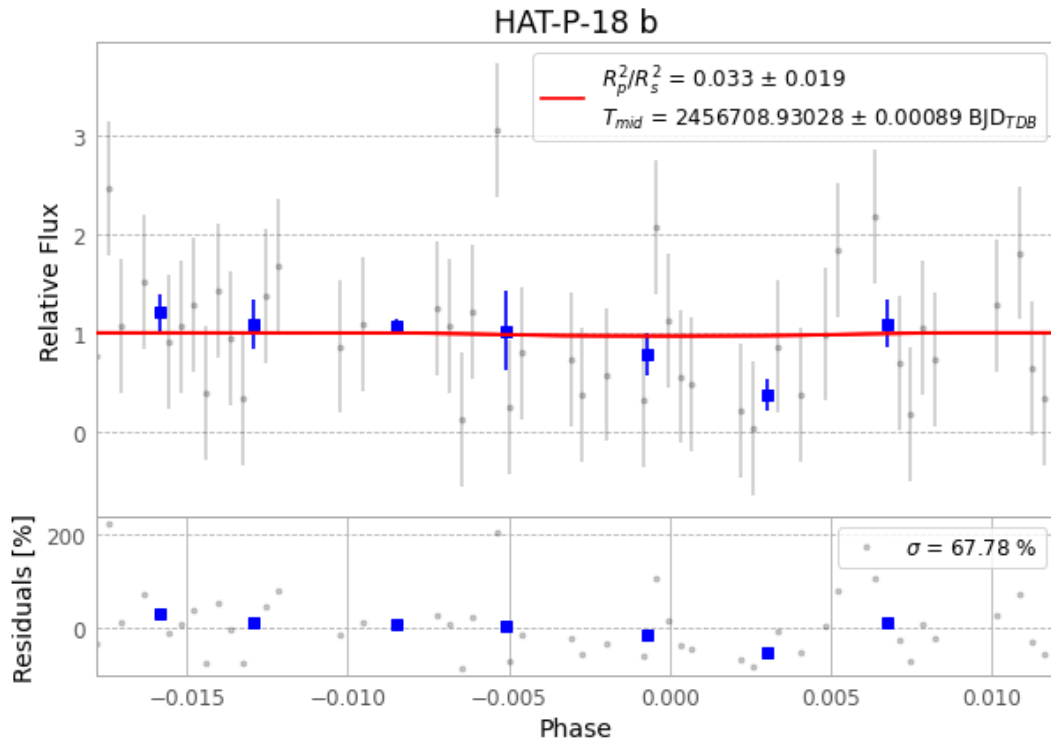


Figure 1 — FinalLightCurve_HAT-P-18 b_2014-02-20.png (SHA-256 2bb6a42e5428a278...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [423, 136], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b57ff66e375c4cd9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

80 FITS frames (53 MB), HATP-18140220075712.FITS → HATP-18140220115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-140220.log (750dd0b99295...), FinalLightCurve_HAT-P-18 b_2014-02-20.png (2bb6a42e5428...), FinalLightCurve_HAT-P-18 b_2014-02-20.csv (addf73c4b18b...)

Compute resources

Wall time 138.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:32Z.